Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9	*(a)*	(i)		Any object [wholly or partially] immersed in a fluid experiences an upwards force equal to the weight of the fluid displaced by the object	1			1		
		(ii)		Sub into $\rho = \frac{m}{V}$ with conversion i.e. 80×10^{-6} and 1 030 (1) Manipulation and answer ($1\,030 \times 80 \times 10^{-6}$) = 0.0824 kg (1)		2		2	2	
		(iii)		Mass of block of ice = 0.0824 kg (can be inferred) (1) Allow **ecf** from (i) Volume of block of ice = $\frac{0.0824}{920} = 89.6 \times 10^{-6}$ [m^3] (1) Volume of ice above surface = 9.6×10^{-6} [m^3] (1)	1	1 1		3	2	
		(iv)	I	$\frac{2.2\times10^{11}\ \text{m}^3\ \text{yr}^{-1}}{3.6\times10^{14}\ \text{m}^2}\left[=6\times10^{-4}\ \text{m yr}^{-1}\right]$		1		1		
			II	Melting ice sheets adds to the volume of the sea (1) Melting icebergs replaces volume of sea water already displaced so water level unchanged / change is negligible (1)		2		2		
			III	More radiation absorbed by darker surfaces that is uncovered or less radiation reflected as white reflective surfaces have melted			1	1		
	(b)	(i)		All units correctly identified: Power = W; [A] = m^2; [$\Delta\theta$] = K; [Δx] = m (1) Convincing algebra e.g. $K = \frac{\Delta Q \Delta x}{\Delta t \Delta \theta A}$ (1)	2			2	1	

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		(ii)	Use of thermal conductivity equation i.e. $\frac{\Delta Q}{\Delta t} = \frac{0.041 \times 72 \times 15}{100 \times 10^{-3}}$ (1) = 443 [J s^{-1} / W] (1)	2			2	2	
		(iii)	$\frac{\Delta Q}{\Delta t}$ the same through both layers (1) $\frac{0.041(20 - \theta)}{100} = \frac{0.035(\theta - 5)}{170}$ seen or boundary temp = 15.0 °C (1) $\frac{\Delta Q}{\Delta t}$ = 148 [W] (1) Accept thermal resistance in series approach i.e. $\frac{15}{\left(\frac{100 \times 10^{-3}}{0.041 \times 72}\right) + \left(\frac{170 \times 10^{-3}}{0.035 \times 72}\right)}$ = 148 [W] > 60% reduction so claim is correct. Allow **ecf** on $\frac{\Delta Q}{\Delta t}$ values(1)			4	4	3	

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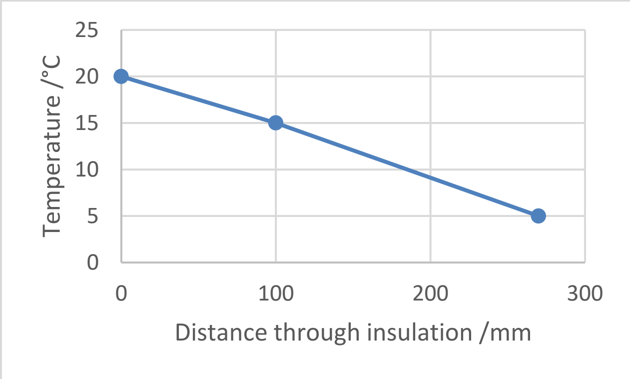
				Marks available					
		(iv)	 Negative linear gradient (1) Correct points at (0,20), (100,15 ecf) and (270,5) (1)		2		2		
			Question 9 total	**6**	**9**	**5**	**20**	**10**	**0**